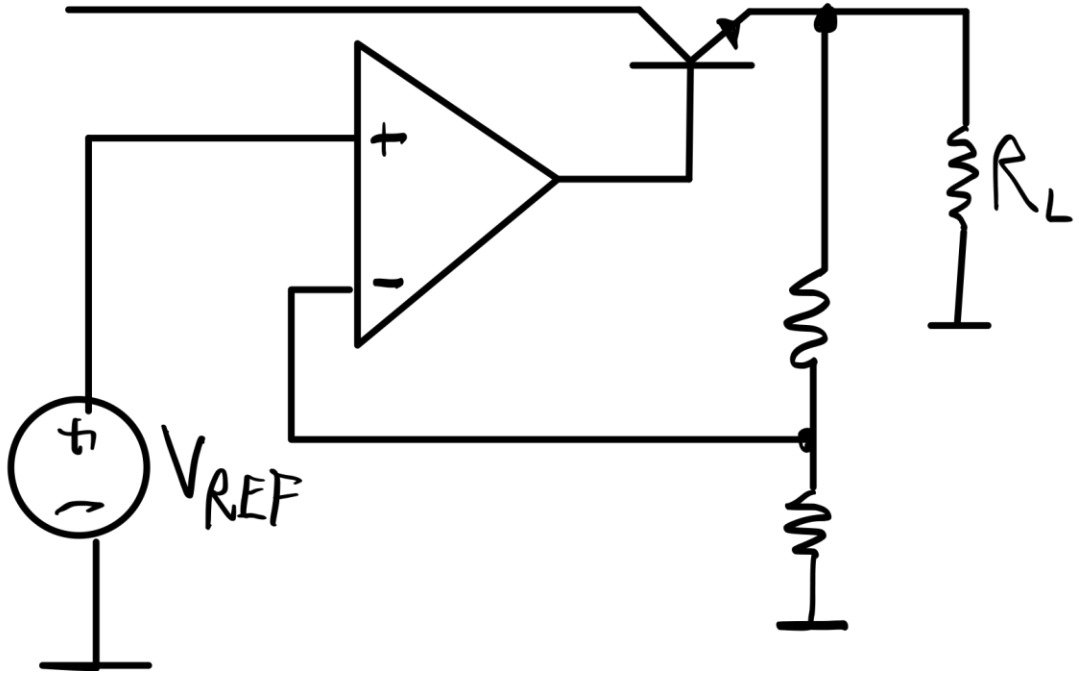
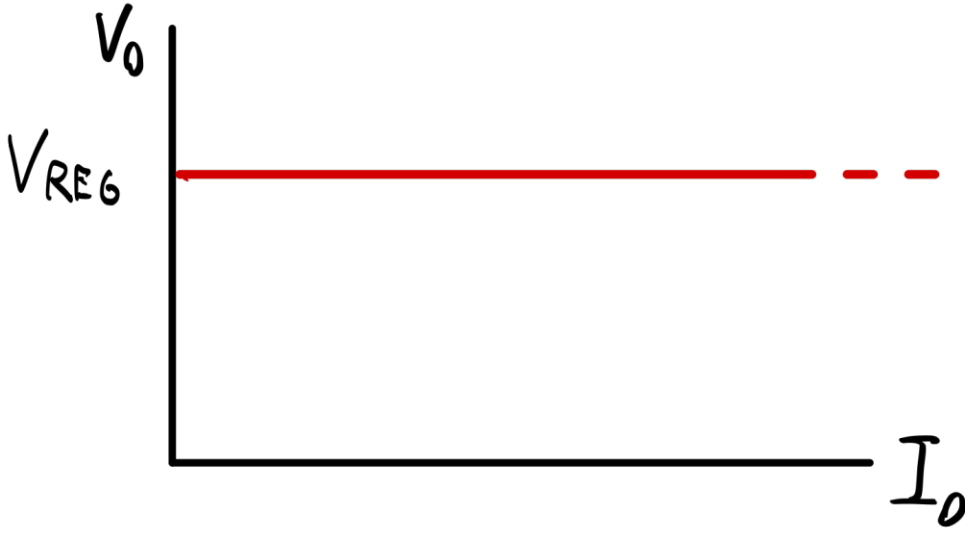
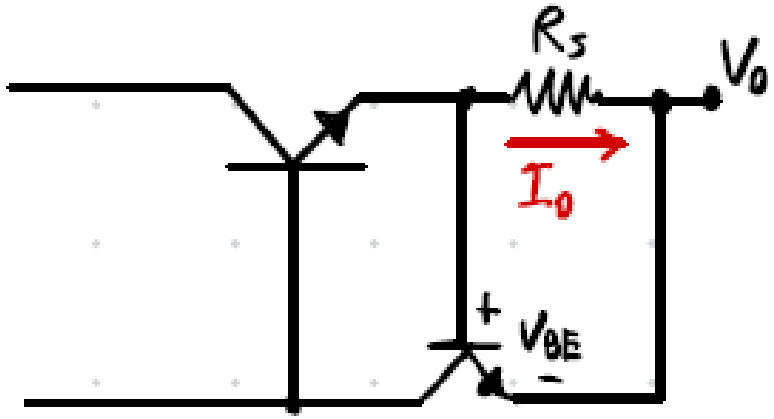
Regulador lineal



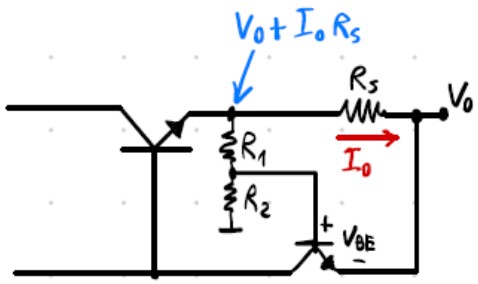


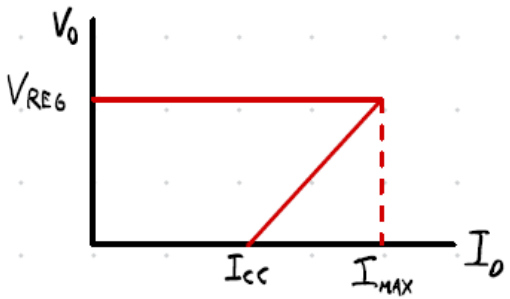
Limitador de corriente



$$I_{MAX} = \frac{V_{BE\,ON}}{R_S}$$

Foldback





$$V_{BE} = V_{RS} - V_{R1}$$

$$V_{BE} = R_S I_0 - (V_0 + R_S I_0)\frac{R_1}{R_1 + R_2} = R_S\left(\frac{R_2}{R_1+R_2}\right) I_0 - \frac{R_1}{R_1+R_2} V_0$$

$$I_0 = \frac{(R_1+R_2)V_{BE} + R_1 V_0}{R_S R_2}$$

$$\boxed{I_0 = \underbrace{\frac{V_{BE}}{R_S}\left(1+\frac{R_1}{R_2}\right)}_{I_{CC}\ (V_0 = 0)} + \frac{V_0}{R_S}\frac{R_1}{R_2}}$$

$I_0 \uparrow \Rightarrow V_{BE} \uparrow$

$\Rightarrow$ T saca corriente de la base

Calculos

$$I_{MAX} = \frac{V_{BEON}}{R_S}\left(1+\frac{R_1}{R_2}\right) + \frac{V_0}{R_S}\frac{R_1}{R_2}$$

$$I_{CC} = \frac{V_{BEON}}{R_S}\left(1+\frac{R_1}{R_2}\right) \longrightarrow R_S = \frac{V_{BEON}}{I_{CC}}\left(1+\frac{R_1}{R_2}\right)$$

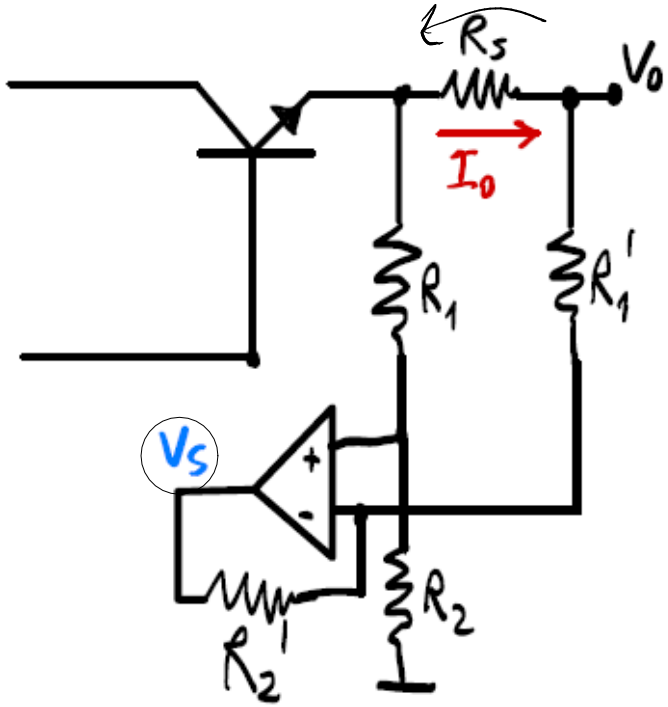
$$\frac{I_{MAX}}{I_{CC}} = 1 + \frac{V_0}{V_{BEON}}\frac{\frac{R_1}{R_2}}{1+\frac{R_1}{R_2}}$$

$$\frac{I_{MAX}}{I_{CC}} = 1 + \frac{V_0}{V_{BEON}}\frac{1}{1+\frac{R_2}{R_1}}$$

$$\left(\frac{I_{MAX}}{I_{CC}}-1\right)\frac{V_{BEON}}{V_0} = \frac{1}{1+\frac{R_2}{R_1}}$$

$$\boxed{\frac{R_2}{R_1} = \frac{V_0}{\left(\frac{I_{MAX}}{I_{CC}}-1\right)V_{BEON}} - 1} \Rightarrow \boxed{R_S = \frac{V_{BEON}}{I_{CC}}\left(1+\frac{R_1}{R_2}\right)}$$

Sensor de corriente

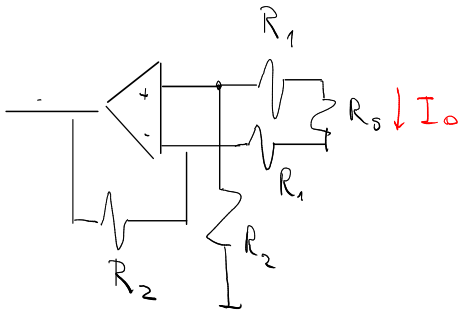


$$A_{VC} = 0 \quad (R_1 = R_1', R_2 = R_2')$$

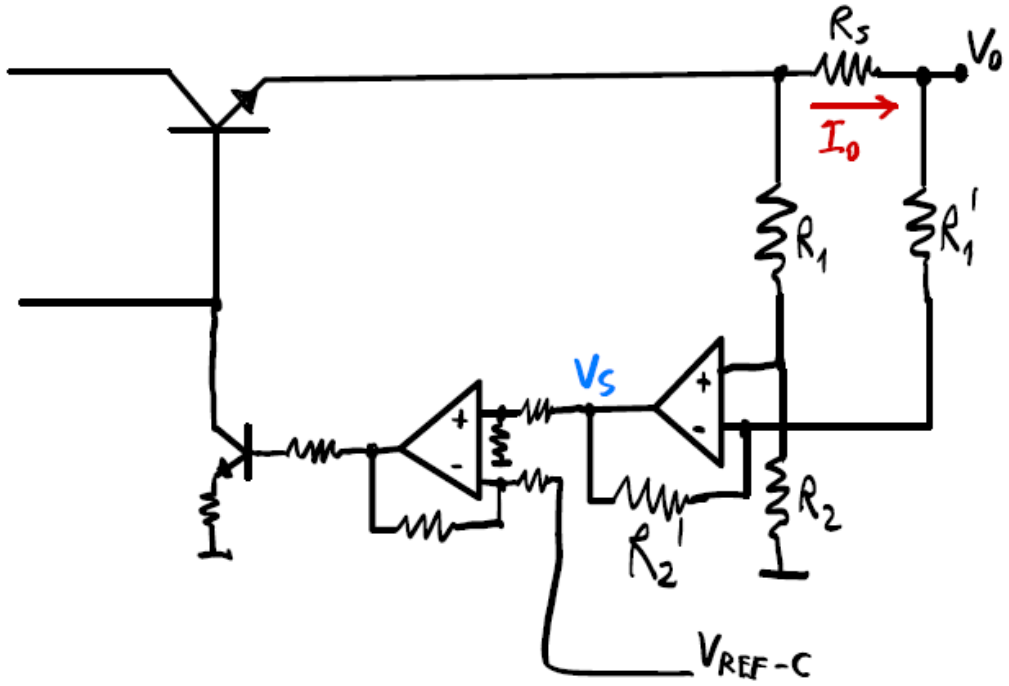
$$A_{VD} = \frac{R_2}{R_1}$$

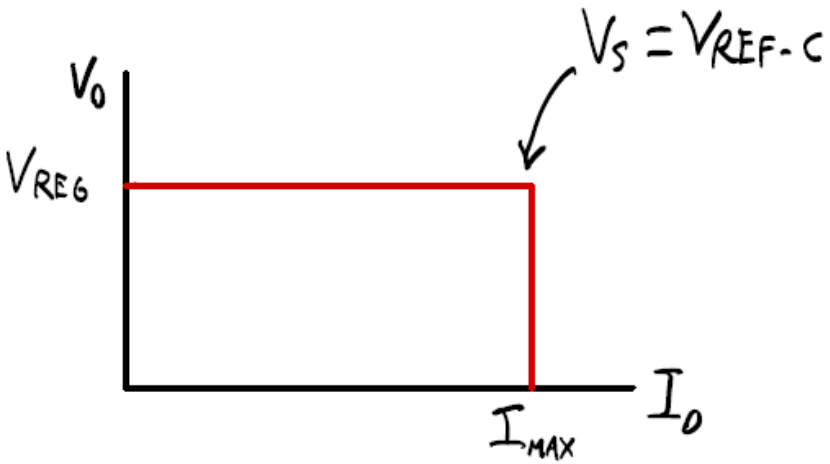
$$\boxed{V_s = \underbrace{I_o \cdot R_s}_{V_{in}} \cdot \frac{R_2}{R_1}}$$

Redibujo



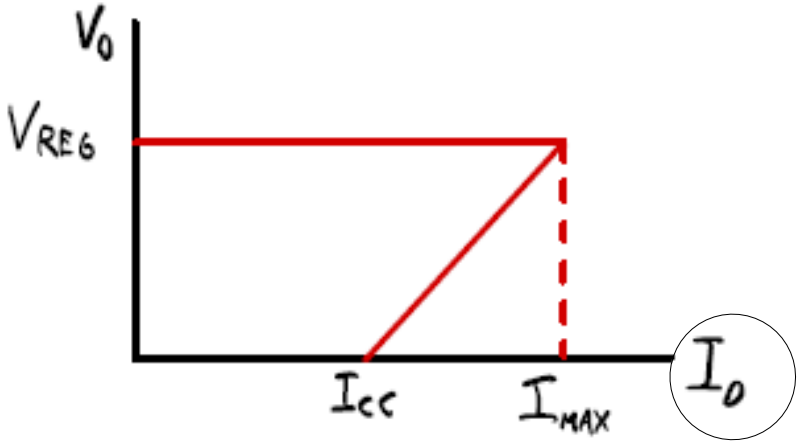
Limitador de corriente





$$V_{REF-C} \simeq I_{MAX} R_s \frac{R_2}{R_1}$$

Foldback



$$V_{REF-C} = \left(I_{cc} + V_0 \cdot \frac{\Delta I}{\Delta V}\right)$$

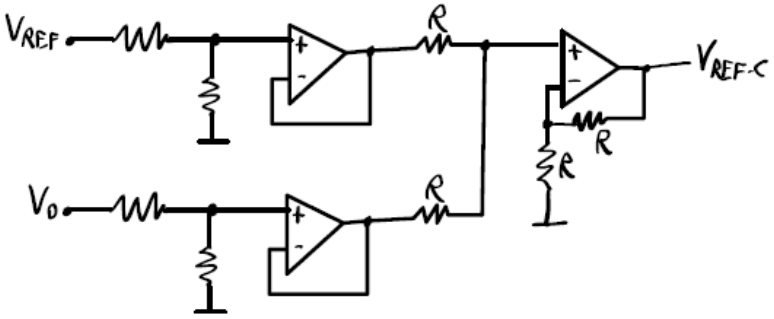
$\Delta V = V_{REG}$

$\Delta I = I_{MAX} - I_{cc}$

$$V_{REF-C} = \underbrace{I_{cc} R_s \frac{R_2}{R_1}}_{\text{FIJO}} + \underbrace{V_0 \cdot \frac{\Delta I}{\Delta V} \cdot R_s \frac{R_2}{R_1}}_{\text{VARIABLE}}$$

$$V_{REF-C} = \alpha V_{REF} + \beta V_0$$

Ejemplo



$V_{REG} = 5V$

$V_{REF} = 2V$

$I_{CC} = 0.5A$

$I_{MAX} = 1A$

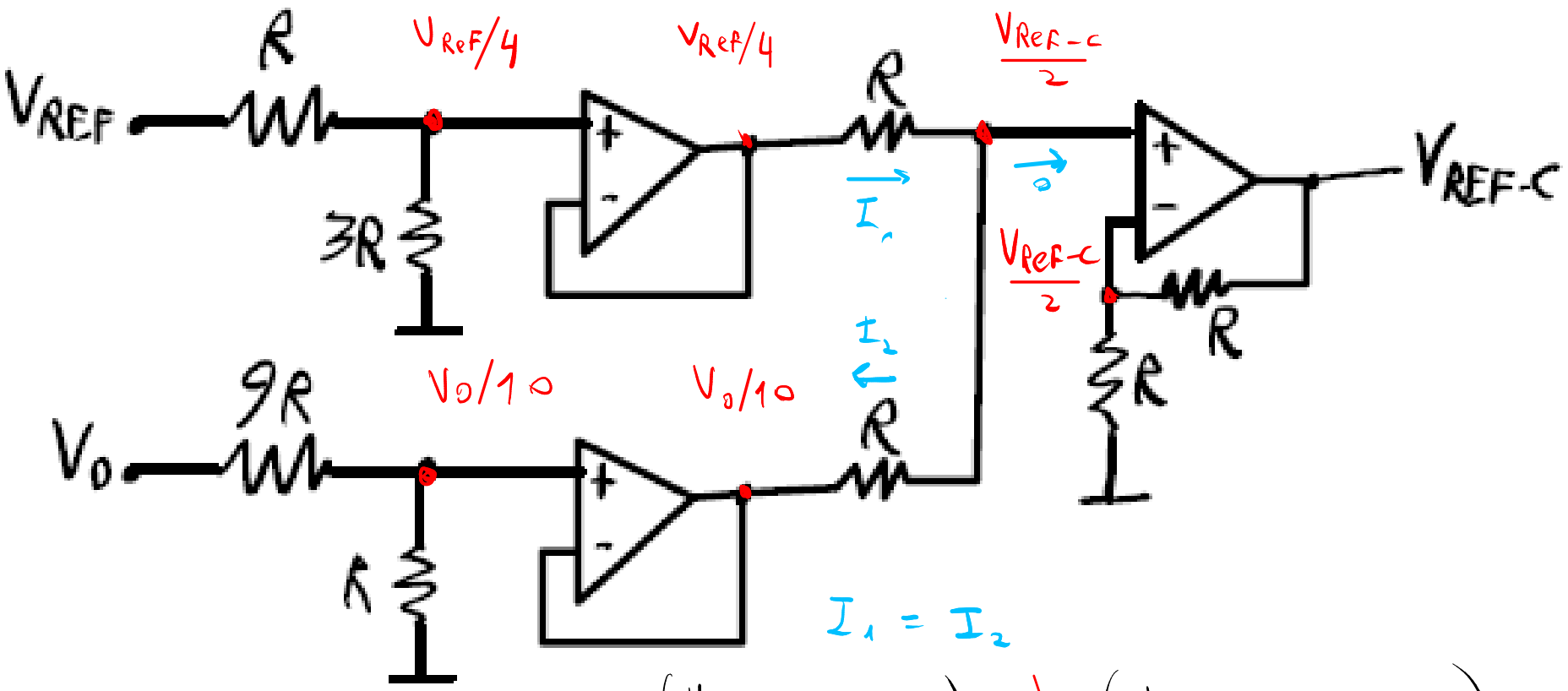
$R_S = 0.1\Omega$

$\frac{R_2}{R_1} = 10$

$$\alpha \cdot V_{REF} = I_{CC} R_S \frac{R_2}{R_1} \Rightarrow \alpha = \frac{I_{CC} R_S R_2}{V_{REF} \cdot R_1} = 1/4$$

$$\beta = \frac{\Delta I}{\Delta V} \cdot R_S \frac{R_2}{R_1} = 0.1$$

Implementacion



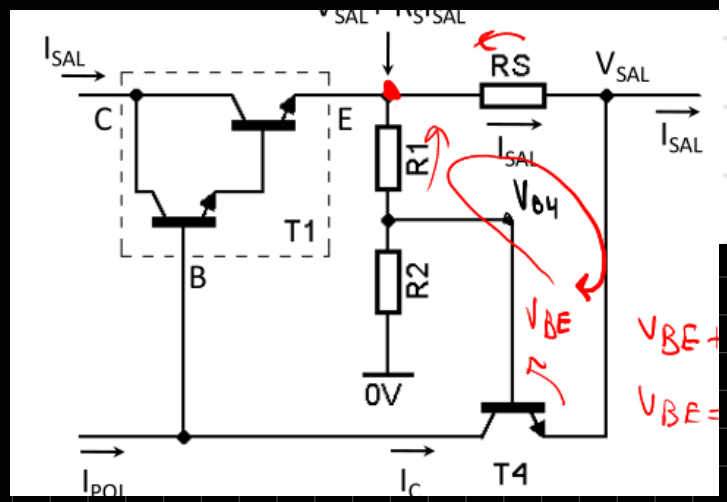
$$I_1 = I_2$$

$$\frac{1}{R}\left(\frac{V_{REF}}{4} - \frac{V_{Ref-c}}{2}\right) = \frac{1}{R}\left(\frac{V_{Ref-c}}{2} - \frac{V_o}{10}\right)$$

$$V_{REF-c} = \frac{V_{REF}}{4} + \frac{V_0}{10} \rightarrow \text{Sumador}$$

Consideraciones

- Se puede implementar con menos operacionales?
- Cuando esta activo el lazo de tension y cuando esta activo el lazo de Corriente?
- Como cambia la implementacion si uso un MOS?



$$\boxed{\frac{R_2}{R_1} = \frac{V_D}{\left(\frac{I_{MAX}}{I_{CC}} - 1\right) V_{BEON}} - 1} \Rightarrow \boxed{R_S = \frac{V_{BEON}}{I_{CC}}\left(1 + \frac{R_1}{R_2}\right)}$$

Requisitos: $I_{CC} = 0{,}4\ A$; $I_{MAX} = 1{,}5$ -

$$\frac{R_2}{R_1} = \frac{5\ V}{\left(\frac{1{,}5}{0{,}4} - 1\right) \cdot 0{,}7\ V} - 1 = \frac{123}{77} \cong 1{,}60 = \frac{8}{5}$$

$$R_S = \frac{0{,}6\ V}{0{,}4\ A} \cdot \left(1 + \frac{5}{8}\right) = \frac{39}{16} \cong 2{,}4$$

Propuesta 1: $R_1 = 5\ K$ y $R_2 = 8\ K$ $\rightarrow R_2/R_1 = 0{,}48 \rightarrow R_S = 5{,}36\ \Omega \rightarrow 2{,}4\ \Omega$

Propuesta 2: $R_1 =$ ___ y $R_2 =$ ___ $\rightarrow R_2/R_1 = 0{,}48 \rightarrow R_S = 5{,}36\ \Omega \rightarrow$ ___

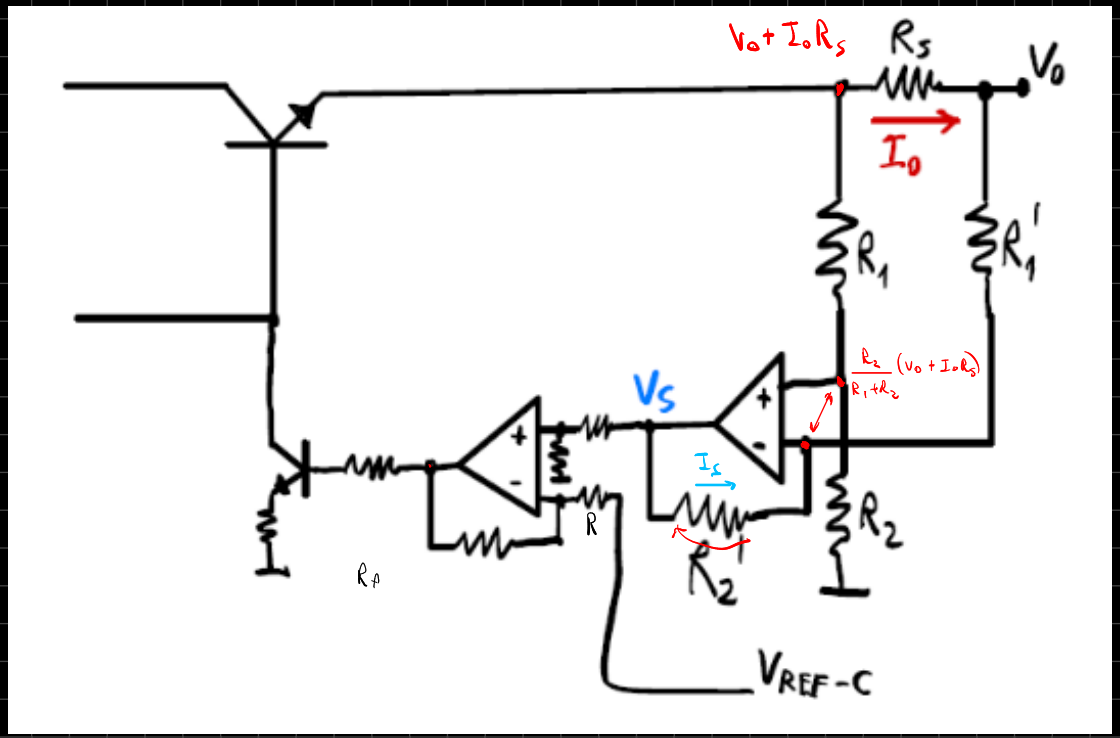
¿Beneficio de q' R_1 y R_2 sean grandes?

¿Desventaja?

$$I_{CC} = \frac{V_{BE4ON}}{R_S}\left(1+\frac{R_1}{R_2}\right) \quad y \quad I_{MAX} = I_{CC} + \frac{R_1}{R_2}\frac{V_{SALMAX}}{R_S}$$

$R_S = 10 \rightarrow R_1/R_2 = R_S \frac{I_{CC}}{V_{ON}} - 1 = 17/3$

Sobre el circuito de Guido:



$$I_S = \frac{V_S - V_0}{R_1' + R_2'}$$

$$\frac{R_2}{R_1+R_2} \cdot (V_0 + I_0 R_S) + I_S R_2'$$

$$= V_S$$

$$\rightarrow \frac{R_2}{R_1+R_2}(V_0 + I_0 R_S) + \frac{R_2'}{R_1' + R_2'}(V_S - V_0) = V_S$$

Busca V_S en función de I_0

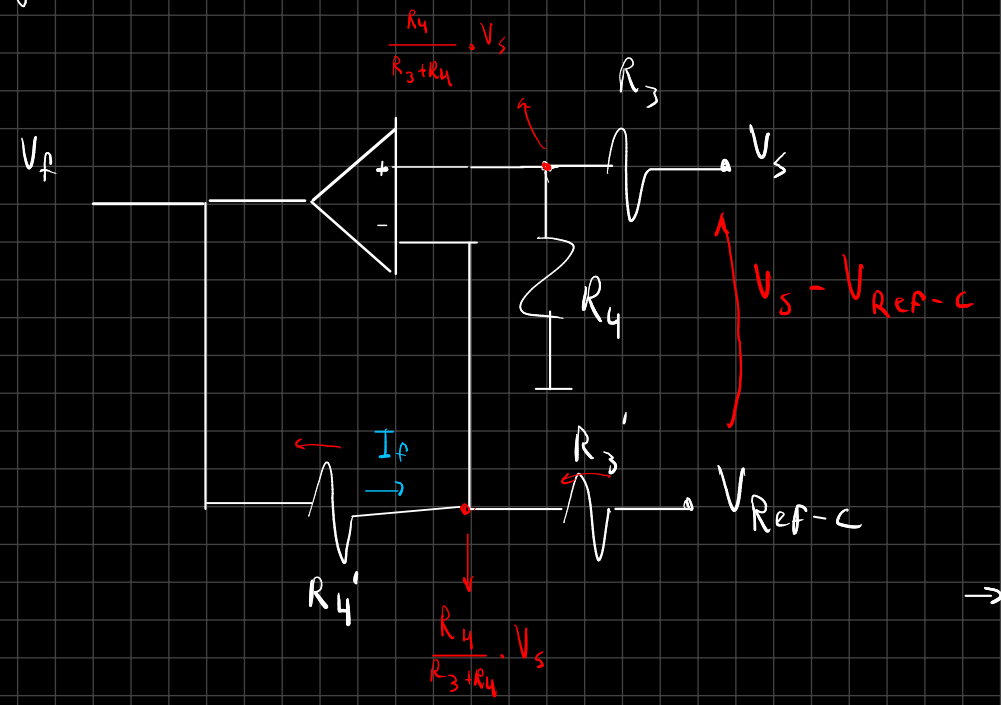
$$(R_1 = R_1',\ R_2 = R_2') \Rightarrow \frac{R_2}{R_1+R_2}(\cancel{V_0} + I_0 R_S + V_S - \cancel{V_0}) = V_S$$

$$\frac{R_2}{\cancel{R_1+R_2}} \cdot I_0 \cdot R_5 = V_5 - \frac{R_2}{R_1+R_2} V_5 = \frac{R_1}{\cancel{R_1+R_2}} V_5$$

$$\Rightarrow V_5 = \frac{R_2}{R_1} \cdot R_5 I_0 \rightarrow \boxed{V_5 = \frac{R_2}{R_1} \cdot R_5 \cdot I_0}$$

La segunda etapa:

Conjetura: $V_f = \frac{R_4}{R_3}\left(V_5 - V_{Ref-c}\right)$



$$I_f = \frac{V_f - V_{Ref-c}}{R_3' + R_4'}$$

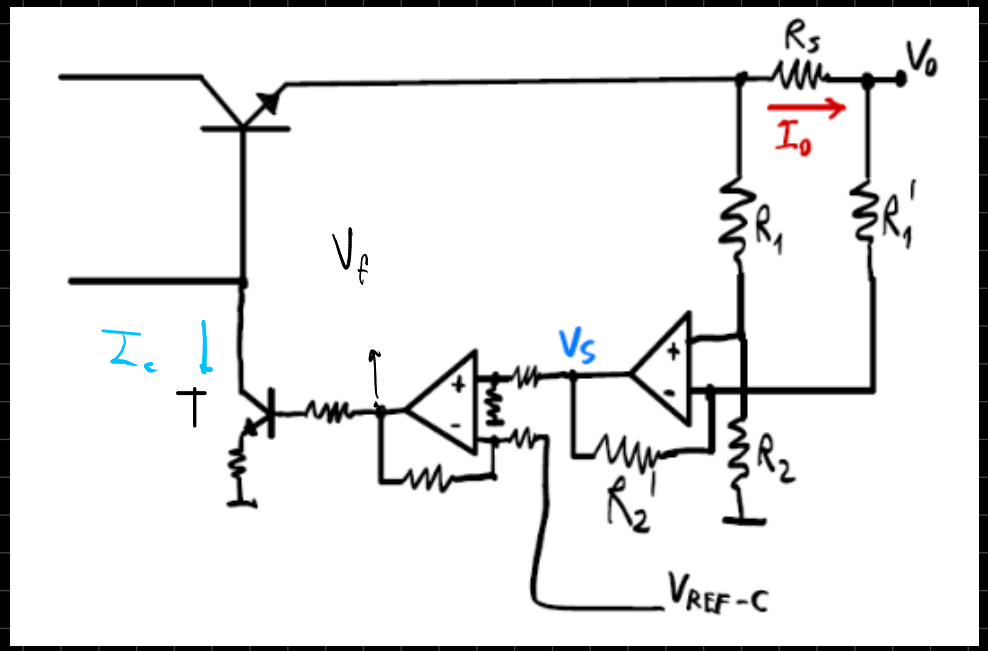
$$V_f = \frac{R_4}{R_3+R_4} \cdot V_5 + R_4' \cdot I_f \quad \left(R_3 = R_3',\ R_4 = R_4'\right)$$

$$\rightarrow V_f = \frac{R_4}{R_3+R_4}\left(V_5 + V_f - V_{Ref-c}\right)$$

$$\rightarrow V_f = \frac{R_4}{R_3+R_4}\left[(V_s - V_{Ref-c}) + V_f\right] \Rightarrow V_f = \frac{R_4}{R_3}\cdot(V_s - V_{Ref-c})$$

$$\therefore \boxed{V_f = \frac{R_4}{R_3}(V_s - V_{Ref-c})}$$

$$V_s \propto I_o$$



$$V_s = \frac{R_2}{R_1}\cdot R_s \cdot I_o$$

$$V_f = \frac{R_4}{R_3}\cdot(V_s - V_{Ref-c})$$

Ejemplo:

i) Si $V_s - V_{Ref-c} < 0 \Rightarrow V_f < 0 \Rightarrow$ T NO conduce

ii) Si $V_s - V_{Ref-c} = 0 \Rightarrow$ T NO conduce

iii) Si $V_s - V_{Ref} = 0{,}07\,V$ y $R_4/R_3 = 10 \Rightarrow V_f = 0{,}7\,V \rightarrow$ T conduce y alterna corriente

En resumen, el circuito funciona así:
1) Sensa Io. Convierte Io en una tensión proporcional: Vs = R2/R1 * Rs * Io
2) Compara esa tensión Vs con una referencia Vref-C
2a) Si Vs es chica (Si Io es chica), la diferencia Vs - Vref-C es negativa y el transistor T no conduce
2b) Cuando Vs = Vref-C, el transistor T empieza a drenar corriente de la base.

¿Cuando quiero q' empiece a drenar? → Cuando $V_S = V_{S\,máx} = \frac{R_2}{R_1} \cdot R_S\, I_{o\,máx}$

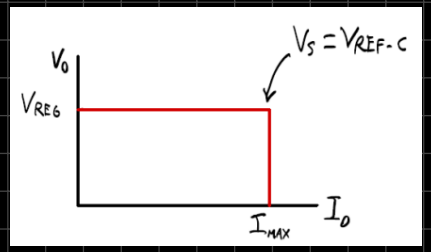
esta ganancia amplifica la diferencia

$\Rightarrow V_{Ref-C} = \frac{R_2}{R_1} R_S \cdot I_{o\,máx}$

$\sim 0{,}01\,V$

→ Cuando $V_S \sim V_{Ref-C} \rightarrow V_f = \boxed{\frac{R_4}{R_3}} (V_S - V_{Ref-C}) \sim 0{,}5\,V$

Este circuito da un <u>LIMITADOR</u> de corriente, <u>NO</u> un foldback.



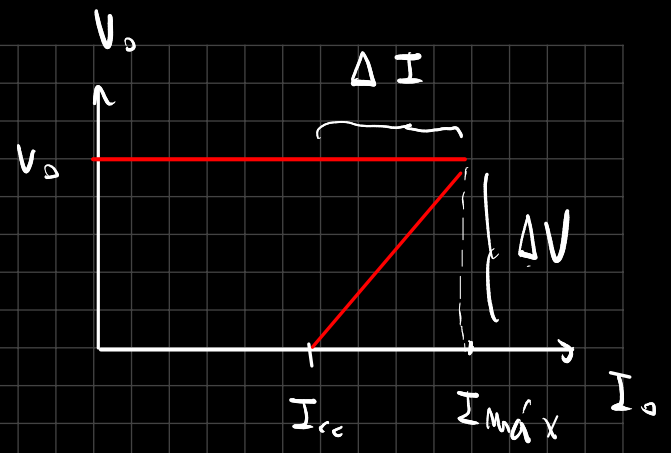
¿Como convierto esto en un foldback? → Tengo q' hacer q la tensión V_{Ref-C} sea <u>variable</u> en lugar de fija

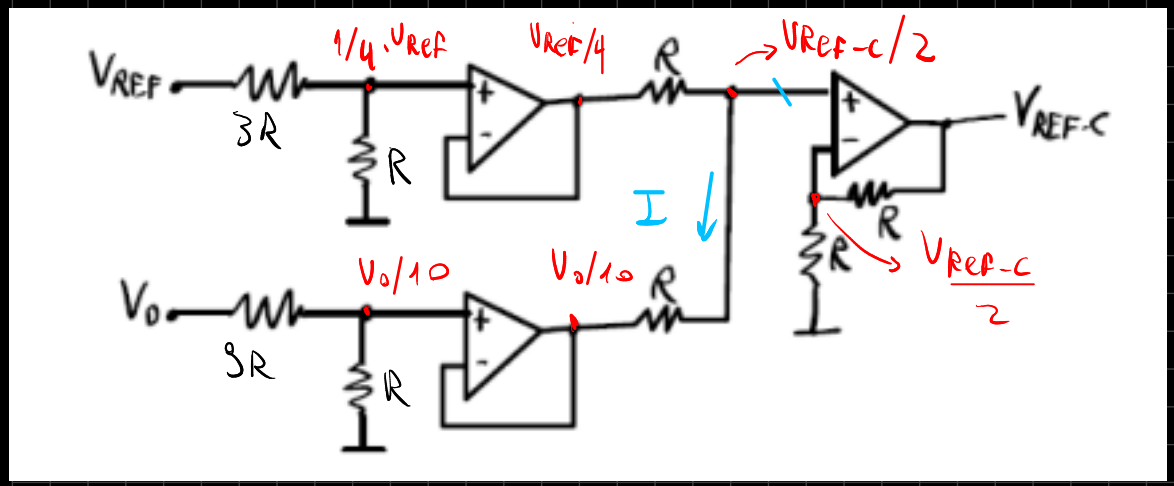
Antes: $V_{Ref-c} = \frac{R_2}{R_1} R_s \overbrace{(I_{máx})}^{CTE} \propto I_{máx}$

Ahora: $V_{Ref-c} = \frac{R_2}{R_1} \cdot R_s \overbrace{\left(I_{cc} + \frac{\Delta I}{\Delta V} \cdot V_o\right)}^{Recta}$

$$= \overbrace{\frac{R_2}{R_1} R_s \cdot I_{cc}}^{CTE\ (O.\ de\ origem)} + \overbrace{\frac{R_2}{R_1} \cdot R_s \cdot \frac{\Delta I}{\Delta V}}^{CTE\ (pendiente)} \cdot V_o$$

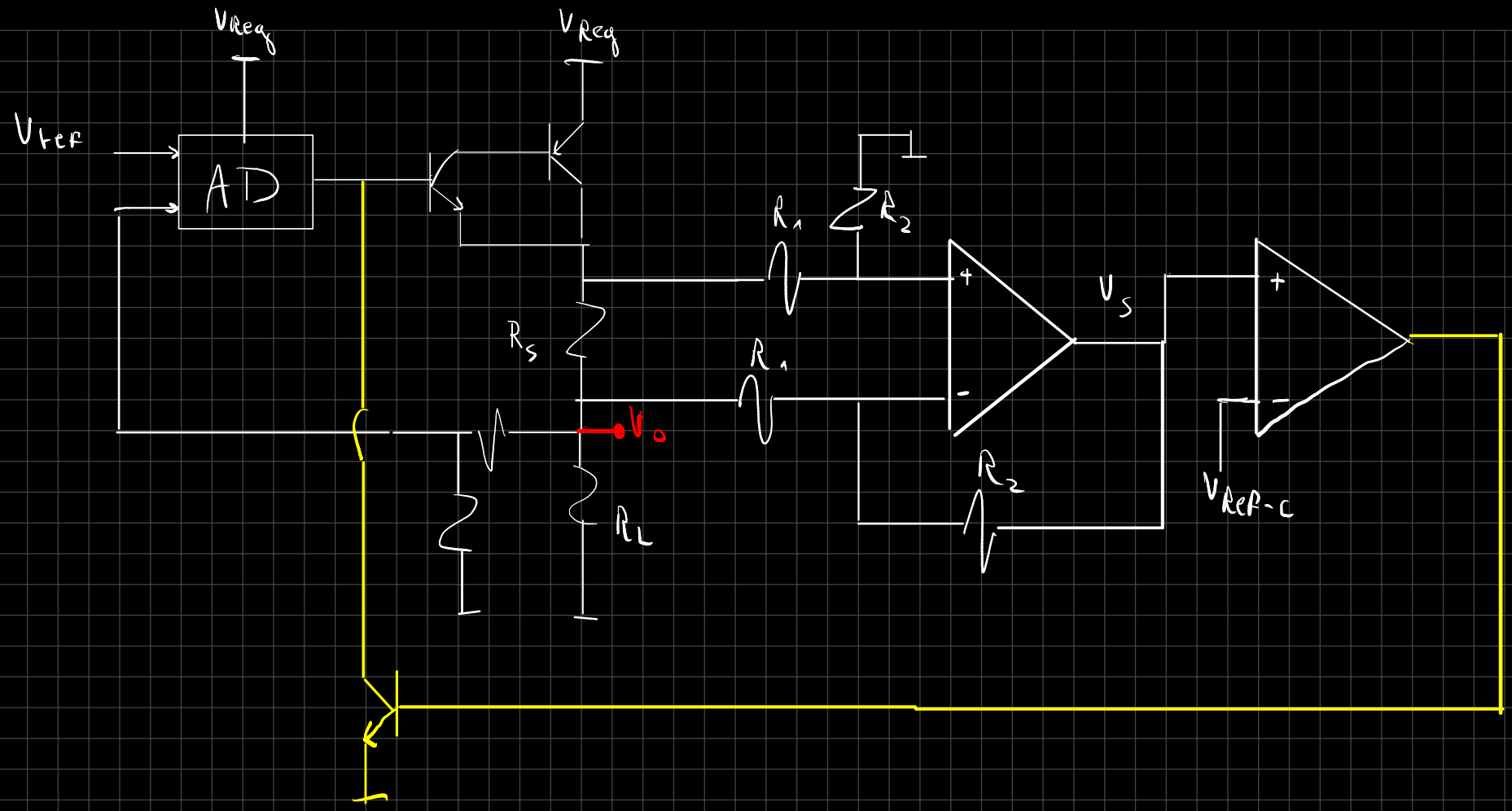
$$= \alpha \cdot V_{Ref} + \beta \cdot V_o$$ → Lo puedo implementar con un sumador



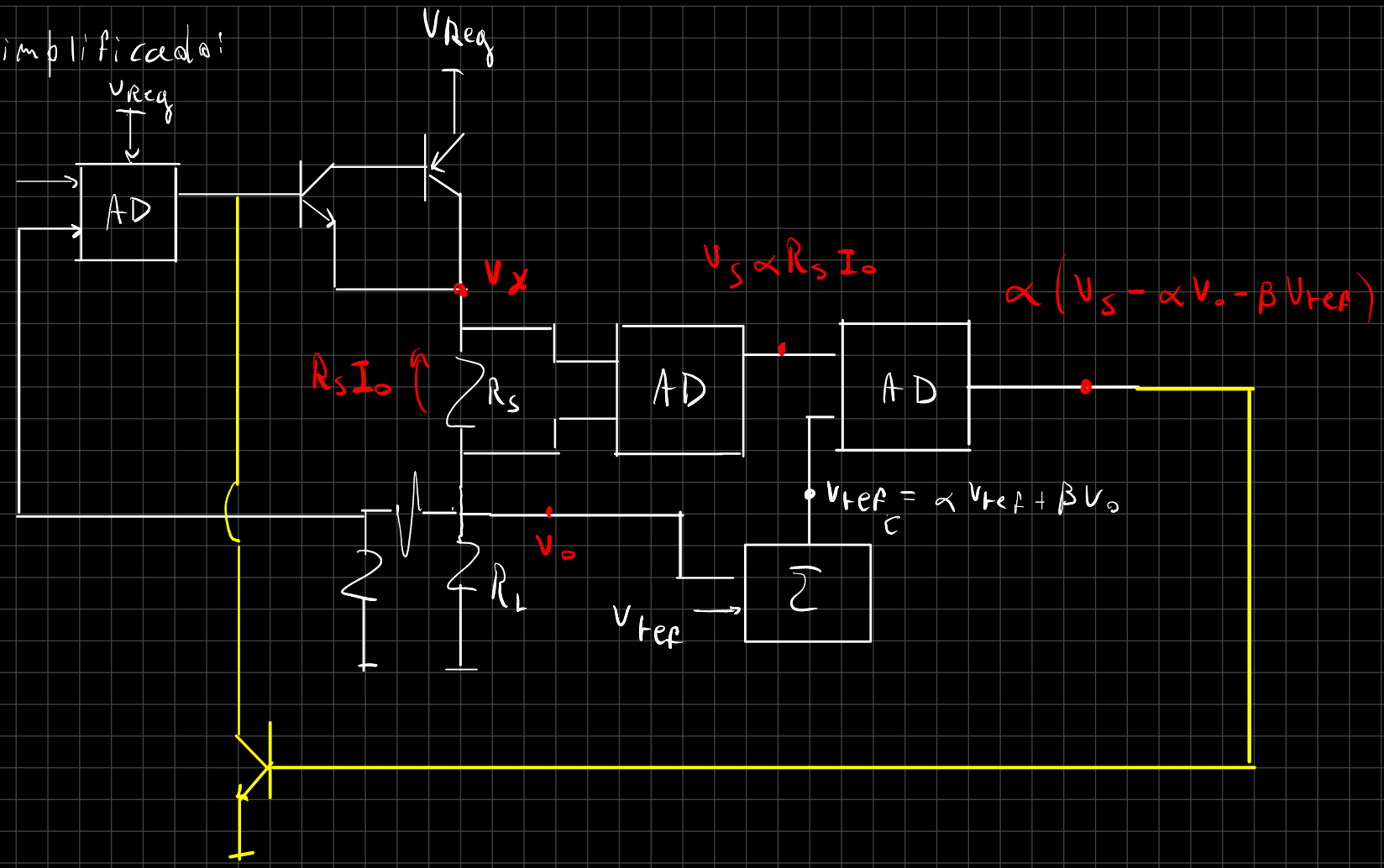


$$I = \frac{1}{R}\left(\frac{V_{Ref-c}}{2} - \frac{V_o}{10}\right) = \frac{1}{R}\left(\frac{V_{Ref}}{4} - \frac{V_{Ref-c}}{2}\right)$$

$$V_{Ref-c} = \underbrace{\frac{1}{4} V_{Ref}}_{O.\ de\ origem} + \underbrace{\boxed{\frac{1}{10}}}_{pendiente} V_o$$


VReq
VReq
Vref
AD
R1
R2
Rs
R1
Vo
R2
RL
Vs
VRef-c

Más simplificado:



Cálculos: Especificaciones: $V_o = 5V$ $\quad I_{o\,máx} = 1,5A$ $\quad \Delta I = 1,1A$ $\quad V_{Ref} = 2,5V$

$\hookrightarrow \Delta V = 5V$ $\quad I_{cc} = 0,4A$

Elijo: $\boxed{R_s = 1\ \Omega}$ $\quad \boxed{R_2/R_1 = 1}$

$$\alpha \cdot V_{Ref} = \frac{R_2}{R_1} \cdot R_s \cdot I_{cc} \rightarrow \alpha = \frac{R_2/R_1 \cdot R_s \cdot I_{cc}}{V_{Ref}} = \frac{0,4}{2,5} = \frac{4}{25} \Rightarrow \boxed{\alpha = 4/25}$$

$$\beta = R_2/R_1 \cdot R_s \cdot \frac{\Delta I}{\Delta V} = 1 \cdot \frac{1,1A}{5V} = \frac{11}{50} = 0,22 \Rightarrow \boxed{\beta = 11/50}$$

$$\therefore V_{Ref-c} = \frac{4}{25} \cdot V_{Ref} + \frac{11}{50} \cdot V_o$$

Por último, elijo $R_4/R_3 = 51$

Ecuaciones importantes:

$$V_S = \frac{R_2}{R_1} \cdot R_S \cdot I_0$$

$$V_{Ref-c} = \alpha \cdot V_{Ref} + \beta V_0$$

$$V_f = \frac{R_4}{R_3}\left(V_S - V_{Ref-c}\right)$$

¿Como se puede hacer con un solo OPamp?

$$V_f = \frac{R_4}{R_3} \cdot \left(\frac{R_2}{R_1} \cdot R_S\right) \cdot I_0 - \frac{R_4}{R_3} \alpha V_{Ref} - \frac{R_4}{R_3} \cdot \beta V_0 =$$

$$= a' \cdot V_{R_S} + b' \cdot V_{Ref} + c' \cdot V_0 = \left(V_{R_S} = V_x - V_0\right)$$

$$= a' V_x + b' \cdot V_{Ref} + (c' - a') V_0 =$$

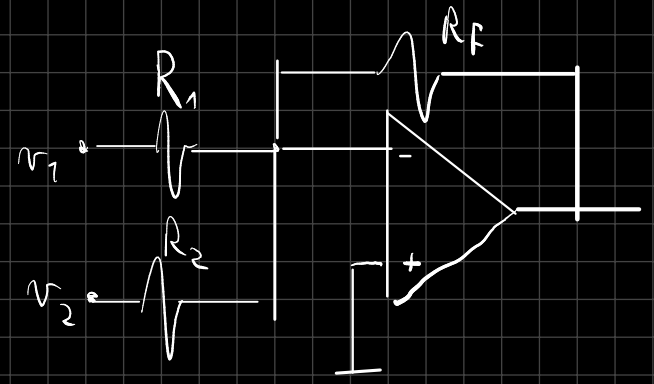
$$= a V_x + b V_{Ref} + c \cdot V_0$$

$$a = \frac{R_4}{R_3} \frac{R_2}{R_1}$$

$$b = -\frac{R_4}{R_3} \cdot \alpha$$

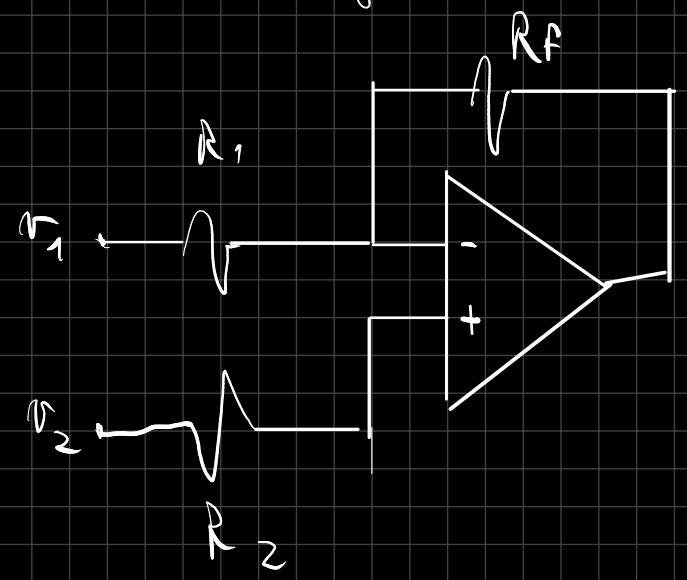
$$c = -\frac{R_4}{R_3} \beta - \frac{R_4}{R_3} \frac{R_2}{R_1} = -\frac{R_4}{R_3}\left(\beta + \frac{R_2}{R_1}\right)$$

Sumador (inversor)



$$v_o = -\frac{R_f}{R_1} v_1 - \frac{R_f}{R_2} v_2$$

<u>Sumador</u> (con y sin inversión)



$$v_o\Big|_{v_2=0} = -\frac{R_f}{R_1} v_1$$

$$v_o\Big|_{v_1=0} = \left(1 + \frac{R_f}{R_1}\right) v_2$$

$$\Rightarrow v_o = -\frac{R_f}{R_1} v_1 + \left(1 + \frac{R_f}{R_1}\right) v_2$$

Em geral: $v_o = \underbrace{-\sum_i \frac{R_f}{R_i} \cdot v_i}_{\text{entradas inversoras}} + \underbrace{\sum_j \left(1 + \frac{R_f}{R_j}\right) \cdot v_j}_{\text{entradas não inversoras}}$

Cálculos:

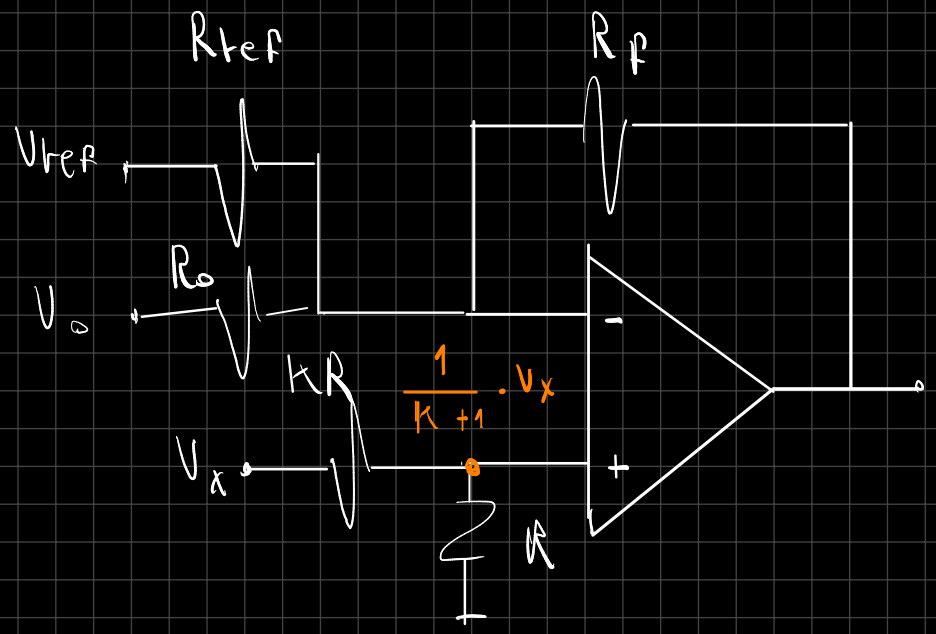
$$a = \frac{R_4}{R_3} \frac{R_2}{R_1} = 51 \cdot 1 = 51$$

$$b = -\frac{R_4}{R_3} \cdot \alpha = -51 \cdot \frac{4}{25} = -\frac{204}{25} = -8{,}16$$

$$c = -\frac{R_4}{R_3}\beta - \frac{R_4}{R_3}\frac{R_2}{R_1} = -\frac{R_4}{R_3}\left(\beta + \frac{R_2}{R_1}\right) = -51\left(\frac{11}{50} + 1\right) = -\frac{3111}{50} = -62{,}22$$

$$V_p = \overset{a}{51} \cdot V_x + \overset{b}{\left(-\frac{204}{25}\right)} V_{ref} + \overset{c}{\left(-\frac{3111}{50}\right)} \cdot V_o$$

Circuito a implementar



$$V_f = -\frac{R_f}{R_{ref}} \cdot V_{ref} - \frac{R_f}{R_o} V_o + \left(1 + \frac{R_f}{R_{ref}//R_o}\right)\frac{1}{k+1} \cdot V_x$$

$$\begin{cases} a = \left(1 + \dfrac{R_f}{R_{ref}//R_o}\right)\dfrac{1}{k+1} = 51 \\ b = -\dfrac{R_f}{R_{ref}} = -\dfrac{204}{25} \\ c = -\dfrac{R_f}{R_o} = -\dfrac{3111}{50} \end{cases} \Rightarrow \begin{cases} R_f = [51k + 50] \cdot R_{ref}//R_o \\ R_f = \dfrac{204}{25} R_{ref} \\ R_f = \dfrac{3111}{50} \cdot R_o \end{cases}$$

Propuesta: $R_f = 10k \Rightarrow k = \frac{1019}{2550}$; $R_{ref} = 1{,}23k$; $R_o = 160{,}7\,\Omega$